

Why Opus 4.5 Solves CORE-Bench

Opus 4.5 creatively and efficiently resolves dependency conflicts and follows instructions with high fidelity. Opus 4.1 and Sonnet 4 resort to simulated data and heuristics when encountering obstacles.

Task Topic	Opus 4.1	Sonnet 4	Opus 4.5
Measuring contagious disease spread	Hit dependency issues and did not run script	Hit dependency issues and resorted to hard coded values	Fixed dependencies, ran intended workflow
Computing response rates	Hit dependency issues, wrote incorrect ad hoc script	Correct values, wrong precision	Rendered files, cross-checked logic
Counting methods counter-arguments	Misinterpreted the question	Used incorrect summary statistic	Targeted correct metric
Twitter bot detection with ML	Used simplified scripts with incorrect parameters	Hit dependency issues, used incorrect metric	Patched code and extracted metrics
Comparing temp prediction models	Hit dependency issues, wrote oversimplified replication scripts	Guessed without inspecting figure	Re-ran notebook, inspected plot
Analyzing plot colors	Accidentally returned an error message as an answer	Analyzed the wrong figure	Installed a missing dependency
Drug integration research	Hit dependency issues, misinterpreted plot	Hit dependency issues, used general knowledge estimate	Fixed bugs, ran end-to-end
Political party voting shares	Ran out of disk space	Ran out of disk space, guessed answer	Installed key dependency
Neurofeedback power analysis	Incorrect precision in answer	Hit dependency issues, manual approximation	Installed key dependency
Epidemic simulation	Hit dependency issues, incorrect custom script	Hit dependency issues, used incorrect simulation	Found compatible version for package

Success factors Failure modes

Table shows qualitative analysis of CORE-Bench capsules where all three models were evaluated with the same Claude Code CLI scaffold.
Green cells summarize success factors; red cells summarize failure modes. Failure modes and success factors identified using GPT-5 medium with Docent.

